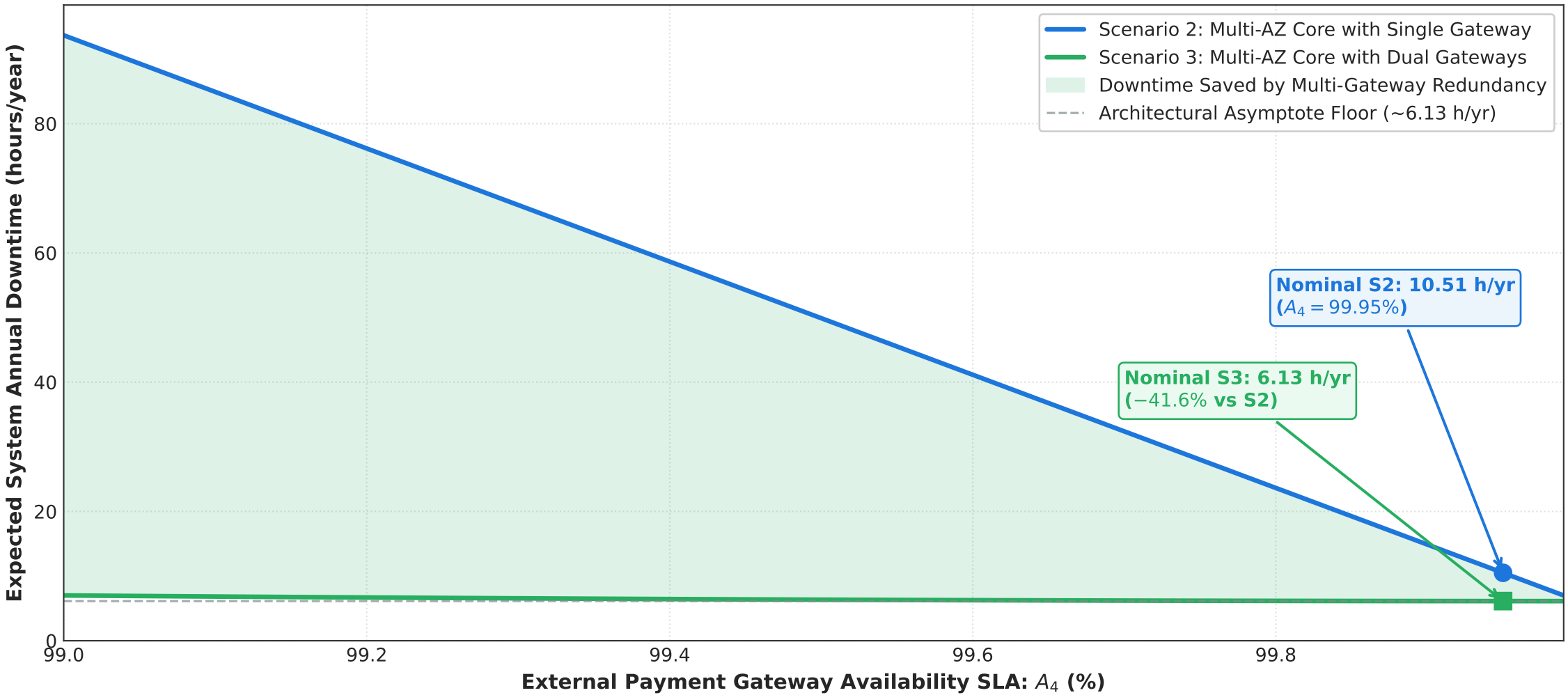
Sensitivity Analysis: System Downtime vs. External Gateway SLA (A_4)
Demonstrating Architectural Shielding Against Third-Party Degradation
Scenario 2: Multi-AZ Core with Single Gateway
Scenario 3: Multi-AZ Core with Dual Gateways
Downtime Saved by Multi-Gateway Redundancy
Architectural Asymptote Floor (~6.13 h/yr)
Nominal S2: 10.51 h/yr
($A_4 = 99.95\%$)
Nominal S3: 6.13 h/yr
(−41.6% vs S2)
Expected System Annual Downtime (hours/year)
0
20
40
60
80
99.0
99.2
99.4
99.6
99.8
External Payment Gateway Availability SLA: A_4 (%)